

Homework Classical Physics, Fall 2023

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Problem Set 04.3 (L07 & L08)

Due on Thursday, 5. October 2023, at 09:30 (in the morning!)

(= Deadline for submitting a pdf file with solution)

Mechanics - Energies (4 points)

A flexible rope is held on a frictionless table with one-third of its length hanging over the edge of the table. If the rope has a length $L = 78$ cm and mass $m = 63$ g (homogeneously distributed along its length). How much work is required to bring the hanging part back on the table by slowly pulling at the end on the table.

- a)** Calculate by using energy conservation.
- b)** Calculate using work and force with $dW = F dx$.

Neglect any effects due to the table edge or bending of the rope.